

Designing Distributed Plastic Recycling Networks

A GIS-Informed Multi-Objective Optimization Framework

Work in progress. Case study: Ville de Nancy, France

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Why the question is territorial

The pressure

- Growing plastic production keeps straining waste systems, and centralized facilities are proving insufficient [1].
- France: **4.5 Mt/year** of plastic waste, only **23% recycled** [2], below the EU **50%** packaging target for 2025 [3].
- Métropole du Grand Nancy: **−15%** per-capita waste, 2010–2030 [4].

Why it is hard

- Waste is generated unevenly across a city; economic, environmental and coverage goals rarely favor the same layout [5].
- Local accessibility and operational constraints govern whether a configuration is viable at all [6, 7].

Literature (I): distributed plastic recycling

An emerging alternative

- Circular-economy thinking has pushed attention toward distributed alternatives [8].
- Distributed recycling via additive manufacturing (DRAM): recycled material entering the 3D-printing chain [9].
- Demonstrated on many streams: plastic blends, PETG, batteries, pouches, bags [10–14].

Open question: *where* should it happen?

- Green Fab Labs show economic potential [15]; in-home schemes have also been proposed [16].
- Logistics view: closed-loop network models [17], mobile-unit location [18], process optimization [19].
- But case studies stay **local** [17, 20] and financial viability vs. centralized systems is **still contested** [21, 22].

Literature (II): GIS in waste management planning

A well-established instrument

GIS integrates heterogeneous spatial data, supports siting and strengthens decisions across the collection cycle [23, 24].

Mostly operational

- LCA + GIS to compare collection scenarios [25].
- Selective-collection scenarios and informal recyclers [26].
- 3D routing under fuel consumption [27]; real routing data [28].

Moving toward design decisions

- Location, size and accessibility of infrastructure [29, 30].
- Sorted-waste collection as a *multiobjective spatial* problem: cost vs. walking distance [31].
- MCDA and expert weighting for container siting [32]; location–routing under economic, environmental and social criteria [33]; transfer stations at municipal scale [34].

Yet collection can be organized in structurally different ways (curbside, drop-off, buy-back, deposit–refund) [35], and that choice shapes participation and recovery [36, 37].

The gap we are working on

The pattern across both bodies

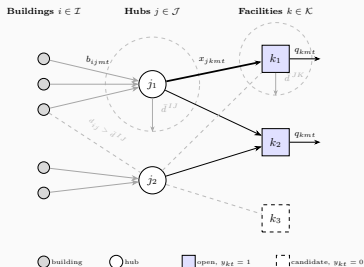
Whether the focus is operational efficiency, design-stage siting or multiobjective spatial planning, GIS is used to **improve a collection scheme already chosen**, not to decide which scheme to adopt.

This matters most for distributed recycling, where spatial conditions strongly shape viability and the comparison against centralized systems remains open [22, 38].

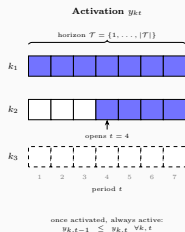
What we are proposing

1. A **two-stage framework**: a reproducible GIS characterization of the territory, feeding a multi-objective optimization model, with spatial and non-spatial parameters kept separate, so it can travel to other cities.
2. Three design dimensions treated **jointly**: economic return, life-cycle GHG, population-weighted coverage.
3. A first demonstration on the **Ville de Nancy**.

The system and the framework



(a) Network topology and arc eligibility



(b) Facility activation over time

Three-node reverse logistics network: buildings $\rightarrow$ hubs $\rightarrow$ processing facilities $\rightarrow$ market.

Stage One: GIS

- Building-level waste estimation from public cartography
- Hub and facility screening
- Network distances $\Rightarrow$ eligible arcs

Stage Two: optimization

- Which facilities to open, and when
- Monthly flows over the horizon
- NPV / net GHG / coverage

Solved as a MOMILP with the augmented ε -constraint method [39]: we recover the trade-off surface, not one weighted compromise.

Optimization model

Decision variables

$y_{kt} \in \{0, 1\}$: facility k active in period t

$b_{ijmt} \geq 0$: drop-off, building i to hub j

$x_{jkmt} \geq 0$: shipment, hub j to facility k

$q_{kmt} \geq 0$: recycle leaving facility k

Objectives

Economic: discounted net present value

$$\max Z_1 = \sum_k \pi_k, \quad \pi_k = \text{Rev}_k + \text{Cred}_k - \text{Tr}_k - \text{Proc}_k - \text{Fix}_k$$

Environmental: net life-cycle emissions

$$\min Z_2 = \sum_t (E_t^{\text{tr}} + E_t^{\text{proc}} - E_t^{\text{sub}})$$

Social: population-weighted coverage

$$\max Z_3 = \sum_i \frac{P_i}{\Gamma_i} \sum_j \sum_m \sum_t b_{ijmt}$$

Reading the results: $Z_2 < 0$ means the network is a net sink; Z_3 is in inhabitant-equivalents, not tonnes.

Operational constraints

Resident supply is an upper bound

$$\sum_j b_{ijmt} \leq w_{imt}$$

Hubs hold no inventory, and have a throughput cap

$$\sum_i b_{ijmt} = \sum_k x_{jkmt}, \quad \sum_{i,m} b_{ijmt} \leq \bar{Q}_j$$

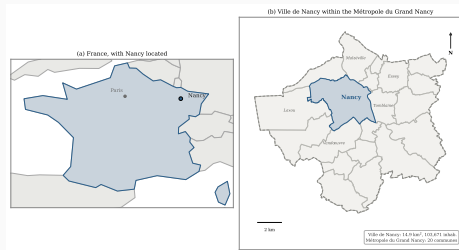
Conversion at polymer yield; the only link between flow and the opening decision

$$q_{kmt} = \eta_m \sum_j x_{jkmt}, \quad \sum_{j,m} x_{jkmt} \leq \bar{Q}_k^{\text{prod}} y_{kt}$$

Opening is irreversible; network size can be capped

$$y_{kt} \geq y_{k,t-1}, \quad \sum_k y_{k|T|} \leq N$$

Case study: Ville de Nancy

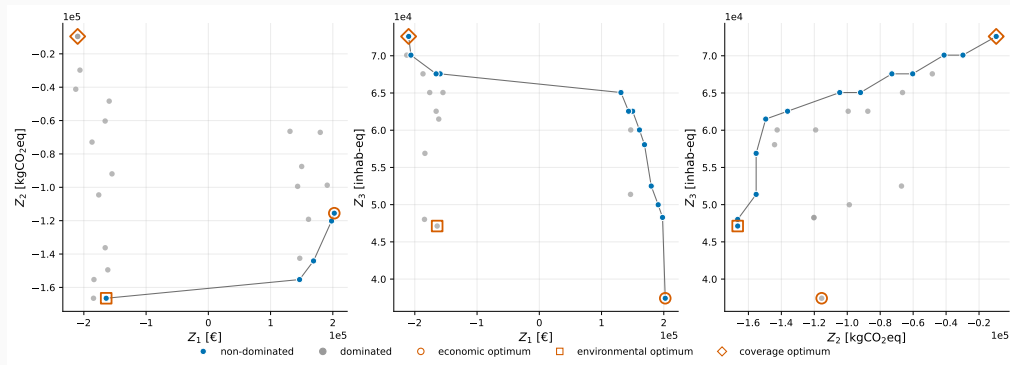


Instance obtained from Stage One

Residential buildings $ \mathcal{I} $	3,111
Collection hubs $ \mathcal{J} $ (schools / retail)	82 (71 / 11)
Candidate processing facilities $ \mathcal{K} $	13
Planning horizon $ \mathcal{T} $	48 months
Polymers	PET, HDPE, PP
Walking catchment $\bar{d}_{ij}$	0.25 km
Operator haul threshold $\bar{d}_{jk}$	1.50 km

Public data only: national building database, urban POI layer, routable road network, which is what makes the characterization reproducible elsewhere.

Preliminary results (I): the trade-off surface



Non-dominated configurations obtained with the augmented ε -constraint method, projected on the three objective pairs.

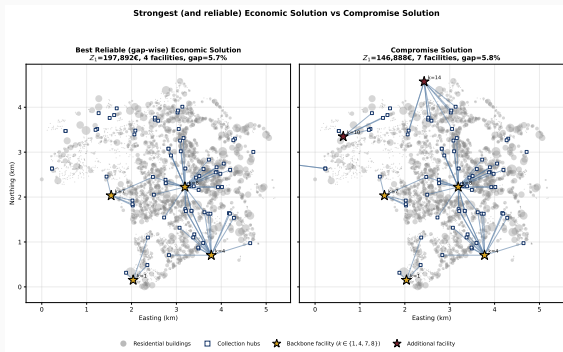
Preliminary results (I): reading the frontier

What we see so far

- A nominal grid of ε values returns **26 mutually non-dominated** configurations.
- Profitability appears to track the **number** of open facilities rather than collected tonnage.
- Every configuration is a **net GHG sink**, but the magnitude varies widely across the set.
- Coverage seems to run into a ceiling well below 100%, which points to an accessibility limit rather than a capacity limit.

Caveat: most subproblems hit the time limit, so this is the best feasible frontier found. We read it qualitatively, not as certified optima.

Preliminary results (II): what the configurations look like



Economic-oriented vs. compromise configuration.

Early observations

- A **stable spatial core** emerges: a few sites open in essentially every configuration.
- The compromise keeps most of the economic return while clearly improving coverage, at the cost of extra facilities.
- The environmental/coverage tension looks **compositional** (which polymers enter) rather than spatial.
- First sensitivity runs: robust to price shocks, more sensitive to catchment and facility scale.

Where this is going

What we think we are learning

- Cardinality, that is, how many facilities to open, may be the authority's main control variable.
- Coverage is bought by making hubs *reachable*, not by enlarging plants.
- Aggregate network profitability can hide individual sites operating at a loss.

Open work

- Tighten the solutions: current optimality gaps limit how strongly we can read the frontier.
- Bring per-site viability into the model explicitly.
- Price and generation uncertainty [40].
- Test transferability in a peri-urban territory, where haul thresholds must be relaxed.

Thank you. Comments are very welcome

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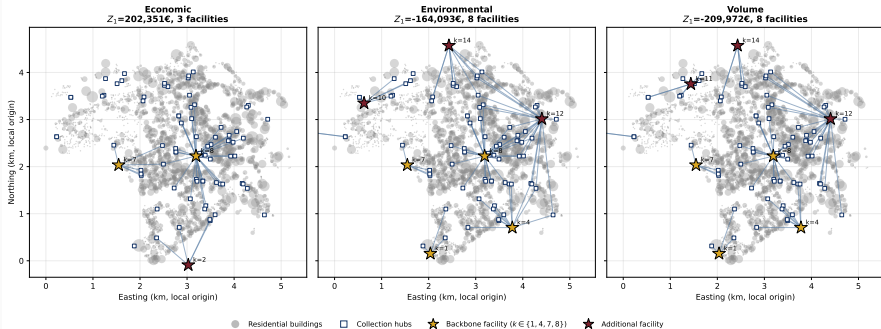
Backup: payoff table

Lexicographic optimum	Z_1 [€]	Z_2 [kgCO ₂ eq]	Z_3	Collected	$ \mathcal{K}^{\text{open}} $
Economic (max Z_1)	202,351	−115,527	37,424	39.7%	3
Environmental (min Z_2)	−164,093	−166,507	47,127	49.9%	8
Coverage (max Z_3)	−209,972	−9,567	72,598	76.9%	8

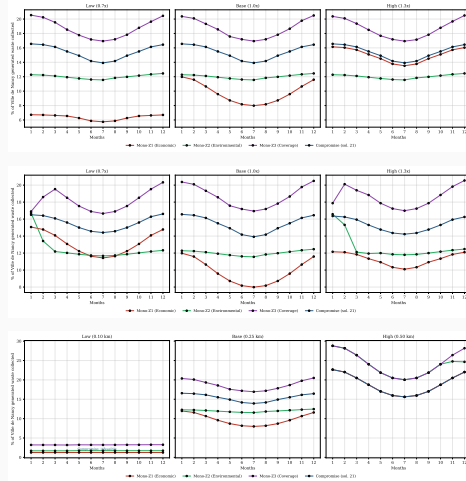
Net balance per kg entering the system: PET −0.36, HDPE −0.04, **PP** +0.07 kgCO₂eq, which is why pushing coverage erodes the environmental objective.

Backup: network of the economic-oriented configuration

Network topology at the three mono-objective solutions



Backup: sensitivity scenarios



Monthly collection under price, facility-scale and catchment-radius scenarios.